Math 12 Curriculum

Intermediate Algebra | Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e

COURSE AT A GLANCE

Review Content

REVIEW

1.1 Use the Language of Algebra; 1.2 Integers; 1.3 Fractions; 1.4 Decimals; 1.5 Properties of Real Numbers

UNIT LEARNING OBJECTIVES

I can evaluate numerical expressions using the order of operations.

I can perform operations with signed integers and fractions.

I can simplify an algebraic expression by combining like terms.

I can use the distributive property to rewrite an expression.

I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

2.1 Use a General Strategy to Solve Linear Equations; 2.3 Solve a Formula for a Specific Variable; 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

UNIT LEARNING OBJECTIVES

I can solve a linear equation and present equivalent steps clearly.

I can solve a linear equation containing fractions.

I can rearrange a formula to isolate a specified variable.

I can translate a verbal relationship into a linear equation.

I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

UNIT LEARNING OBJECTIVES

I can solve and graph a one-variable linear inequality.

I can solve a compound inequality involving AND.

I can solve a compound inequality involving OR.

I can express an inequality solution using interval notation.

I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 3.5 Relations and Functions; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

I can determine and interpret the slope of a line.

I can graph a line from an equation in a common form.

I can write an equation of a line from given information.

I can determine whether two lines are parallel, perpendicular, or neither.

I can evaluate a function using function notation.

I can determine whether a relation is a function using the vertical line test or input-output data.

I can determine domain and range from a graph or table.

I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

UNIT LEARNING OBJECTIVES

I can interpret the solution of a linear system as an intersection point.

I can solve a two-variable linear system by graphing.

I can solve a two-variable linear system by substitution.

I can solve a two-variable linear system by elimination.

I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

5.2 Properties of Exponents and Scientific Notation; 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

UNIT LEARNING OBJECTIVES

I can simplify expressions using integer exponent rules.

I can add and subtract polynomials.

I can multiply polynomials, including binomials.

I can divide a polynomial by a monomial.

I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

2.7 Solve Absolute Value Inequalities; 3.4 Graph Linear Inequalities in Two Variables; 6.1 Greatest Common Factor and Factor by Grouping; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

I can solve and graph an absolute-value inequality.

I can graph a system of linear inequalities and identify its solution region.

I can factor a quadratic trinomial.

I can identify a relationship as discrete or continuous.